Interpolation in 2-D: The Ideas Behind the Code

Grids, scattered data, and the convex hull

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From curves to surfaces

The task

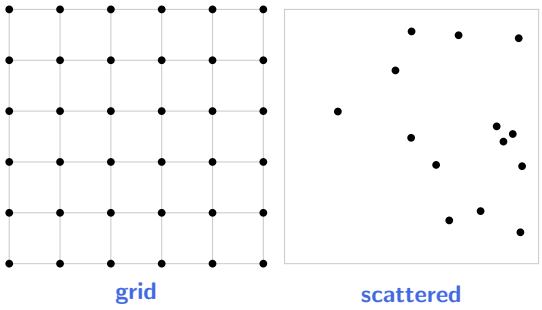
given (x_i, y_i, z_i), find p with $p(x_i, y_i) = z_i$

The data come in **two shapes**:

- **On a grid:** a value at every pair (x_i, y_j).
  tables, images, simulation output
- **Scattered:** values at points with no pattern.
  sensors, weather stations

The first question

Grid or scattered? The answer decides the method.



A grid has **structure** we can use; scattered points do not.

Data on a grid: one direction at a time

A grid point lies in a **rectangle** of data.

Tensor product idea

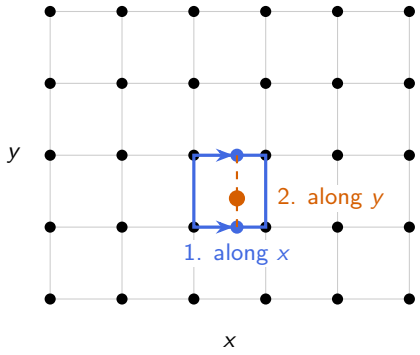
1. Interpolate along x on the grid lines above and below.

2. Interpolate along y between those two values.

Consequence

Every 1-D method carries over: **linear, cubic, PCHIP**.
Each keeps its 1-D character: local or global, safe or smooth.

```
si.RegularGridInterpolator((x, y), Z,
method=...)
```



The grid tells us at once which rectangle a point is in.

Scattered data: triangles and the convex hull

No grid, so we build our own cells: **triangles** with the data points as corners.

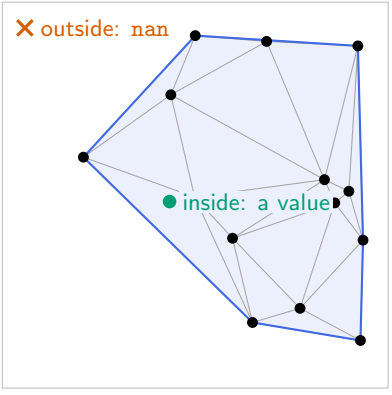
Interpolate on each triangle

A new point lies in one triangle; use its **3 corners** (linear or cubic).
Local: far-away points do not matter.

The convex hull

The triangles cover only the **convex hull**: the smallest convex region holding all the points. Outside it there is no triangle, so no answer (`nan`).

```
si.griddata(pts, z, (Xn, Yn), method=...)
```



Blue outline: the convex hull. Shaded: where we can interpolate.

The big picture

Data	Cells	Where it answers	Tool
grid	rectangles of the grid	the whole grid	`RegularGridInterpolator`
scattered	triangles between points	the convex hull only	`griddata`
scattered	none: one smooth function	everywhere	`RBFInterpolator`

Take-home messages

1. First ask: **grid or scattered?**
2. On a grid, 2-D = **1-D along x, then along y**; the same trade-off: **smooth vs local**.
3. Scattered data live on **triangles**; outside the **convex hull** there is no data.
4. A method that answers outside the data (RBF) is **extrapolating**: trust values near the data.